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05_project / AI / Edge hardware

Off-Grid, AI-Driven Farming

One acre, zero dependency. Two plots 2256 km apart, a solar node that runs on 0.16 Wh a day, and a local model that plans tomorrow morning. No cloud anywhere in the daily loop.

Field node	Data spine	Scheduler
Solar, sealed, 0.16 Wh/day. Sized for a Baltic December.	78 years of station record per plot, provenance kept.	A local LLM proposes. Tested Python disposes.

// the problem

Smart farming assumes a grid that is not there

Most smart farming projects assume mains power, a cloud connection and a budget for commercial sensor kits. This one starts from the opposite constraints: no grid, no runtime network, about €175 of parts, and a decision that still has to be right on the day it is made.

Those are the constraints of any real edge deployment. A remote asset, a link that drops in exactly the weather you built the thing to observe, a model that has to run somewhere cheap, and an answer that gets acted on before anyone has time to check it.

// what it is

One acre, zero dependency

Two plots, 2256 km apart, farmed in opposite halves of the year: **Prora** on the German Baltic coast through summer, **Castelo Branco** in central Portugal through winter. A solar-powered field node measures soil and weather. A laptop-side Python stack merges that with decades of public station and reanalysis records, forecasts its own solar budget, computes the biodynamic day-type from first principles, and hands the lot to a local LLM as the day's plan: which crop, in which bed, in which window.

> How much of a small farm's daily decision-making can genuinely off-grid AI handle end to end, from sensor to scheduler? The repository is built to answer that, not to assume it.

// three commitments

What the project refuses to fudge

Off-grid

No cloud model in the daily loop

The scheduler runs on the laptop against a local model. The network pulls public history during setup and never becomes a runtime dependency of a decision.

Provenance

Every number carries its error bar

A €30 sensor that claims ±0.3 pH does not get to state pH to one decimal. Raw readings stay raw, calibration applies on read, and each channel has a defensible uncertainty before it reaches the model.

Falsifiable

The calendar is a hypothesis, not a rule

The evidence does not support the biodynamic trigon effect. So it is computed correctly, logged as a covariate, and tested against a pre-registered analysis. A null result is a valid output, possibly the most valuable one.

// the engineering call

Why it is not a Raspberry Pi

The brief said Raspberry Pi. A Baltic December said otherwise. A Pi Zero 2 W idles at 0.4 to 0.6 W, which at 54 degrees north means roughly a 60 W panel and 100 Wh of battery, a caravan installation bolted to a birdbox. A duty-cycled Pico 2 W node runs on **0.16 Wh a day, a factor of about 60**, and one LiFePO4 cell drives its VSYS directly: no boost stage, no boost-converter quiescent current.

The trap sits upstream of the processor. A charge controller's own quiescent draw can exceed a well designed node's entire consumption, by 15 to 45 times in the obvious off-the-shelf case. Pick the power path first.

// data honesty

Three sensors that do not measure what the box says

Cheap 7-in-1 soil probes report nitrogen, phosphorus and potassium in mg/kg. The vendor's own literature says the device measures bulk conductivity and multiplies it by three fixed constants, so the three channels are scaled copies of one number. In-soil pH shifts by about 1.5 units between dry soil and field capacity. Capacitive moisture is a relative index, not volumetric water content. None of that stops the build: it sets what each channel is allowed to claim.

// what it is a proof of

The case study is agricultural. The problems are not.

Edge hardware that survives a winter unattended. A local model with a deterministic constraint layer after it that can veto or defer. A pipeline that fails loudly rather than return something plausible. The same three problems as industrial telemetry, remote monitoring, and any AI feature that has to be right rather than convincing.

// stack

MicroPython · Pico 2 W · BLE / LoRa · Python · SQLite · Skyfield · DWD / Open-Meteo · LSTM · Local LLM · Schema-validated JSON · ruff + pytest CI

// status

Phase 0 of ten. What is built is the plan.

Each with a done-when you can check: fourteen unattended days outdoors, ten bucket tips reading exactly ten, seven dated briefs traceable to their inputs. All of it in the README.

Follow it being built

README	github.com/nicomeihuizen-byte/AI-Homestead
Build plan	docs/BUILD_PLAN.md · docs/DECISIONS.md
Contact	meihuizen.ai · linkedin.com/in/nico-meihuizen